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STIDK3013 – RESEARCH METHOD IN IT

Assignment 3 — Research Findings Report (25%)

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Abstract

The spread of social media has greatly made university students vulnerable to cybercrimes namely phishing, scam and social engineering within Facebook. In this study, the author will examine awareness, risk perception, and protection behaviour in students of the Universiti Utara Malaysia (UUM). The study is based on the Protection Motivation Theory (PMT) and its variables influence the reaction of the students to cyber threats, which include perceived severity, vulnerability, and self-efficacy. A cross-sectional survey was carried out using a quantitative approach on a sample of 55 respondents (mainly undergraduates of UUM between the ages 18-23). The results indicate that student behavior is paradoxical: the desired result is to use the service every day and report a high level of a proactive measuressuch as blocking senders (90.9%) and ignoring requests to be a friend, there could be a severe lack of technical self-efficacy. It is worth noting that 69.1 percent of students evaluated their awareness regarding Facebook settings security as low(1 out of 5) and 67.3 percent acknowledged having trouble to differentiate between authentic and deceptive notification. Besides, content validity analysis (CVI), ensured the strength of the research instrument employed. The research finds out that although UUM students have a pre-existing fear of financial loss and identity stealing, they lack the technical confidence to engage in the use of advanced security setting adapting to avoidance tactics. This can indicate that that the university should focus on particular educational interventions, and social media platforms should change their user interface safety requirements.

Keywords: *Phishing, Social Engineering, Protection Motivation Theory, Cybersecurity Awareness, UUM Students.*

Abstrak

Penyebaran media sosial telah banyak menyebabkan pelajar universiti terdedah kepada jenayah siber iaitu pancingan data, penipuan dan kejuruteraan sosial dalam Facebook. Dalam kajian ini, penulis akan mengkaji kesedaran, persepsi risiko, dan tingkah laku perlindungan dalam kalangan pelajar Universiti Utara Malaysia (UUM). Kajian ini berdasarkan Teori Motivasi Perlindungan (PMT) dan pembolehubahnya mempengaruhi reaksi pelajar terhadap ancaman siber, yang merangkumi tahap keterukan yang dirasakan, kerentanan, dan keberkesanan kendiri. Satu tinjauan keratan rentas telah dijalankan menggunakan pendekatan kuantitatif ke atas sampel 55 responden (terutamanya mahasiswa UUM berumur antara 18-23 tahun). Keputusan menunjukkan bahawa tingkah laku pelajar adalah paradoks: hasil yang diinginkan adalah menggunakan perkhidmatan ini setiap hari dan melaporkan tahap langkah proaktif yang tinggi seperti menyekat penghantar (90.9%) dan mengabaikan permintaan untuk menjadi rakan, mungkin terdapat kekurangan keberkesanan kendiri teknikal yang teruk. Perlu diingatkan bahawa 69.1 peratus pelajar menilai kesedaran mereka mengenai keselamatan tetapan Facebook sebagai rendah (1 daripada 5) dan 67.3 peratus mengakui menghadapi masalah untuk membezakan antara pemberitahuan yang sahih dan yang mengelirukan. Selain itu, analisis kesahan kandungan (CVI) memastikan kekuatan instrumen kajian yang digunakan. Kajian ini mendapati bahawa walaupun pelajar UUM mempunyai ketakutan sedia ada terhadap kerugian kewangan

dan kecurian identiti, mereka kekurangan keyakinan teknikal untuk terlibat dalam penggunaan tetapan keselamatan lanjutan yang menyesuaikan diri dengan taktik mengelakkan. Ini menunjukkan bahawa universiti harus memberi tumpuan kepada intervensi pendidikan tertentu, dan platform media sosial harus mengubah keperluan keselamatan antara muka pengguna mereka.

Kata kunci: *Pancingan Data, Kejuruteraan Sosial, Teori Motivasi Perlindungan, Kesedaran Keselamatan Siber, Pelajar UUM.*

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To begin with, we would like to mention that we would have been extremely grateful to our research supervisor who provided us with invaluable guidance, waited with patience, and gave constructive feedback during the course of the given study. Their knowledge in Research Methods in IT played a significant role in taking the course of this project.

We also want to thank the three experts, Muhammad Farhan bin Ismail, Lee Zhi Xuen and Nur Aisyah binti Rahman, in giving us time and effort to deal with the Content Validity Index (CVI) process of validating our research instrument. Their feedback guaranteed the suitability and usefulness of our questionnaire.

It is worth mentioning that 55 individuals involved in the survey were participants of Universiti Utara Malaysia who were interested in this survey and were eager to take part in it on the voluntary basis. This research would not have been possible without their own (sincere) submission of their experience and personal comments on issues of cybersecurity on Facebook.

Lastly, we would like to thank our families and friends who have been there to support and encourage us in the process of doing this assignment.

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CHAPTER ONE
INTRODUCTION

1.1 Background of the Study

In the current digital era, social media has been involved in an integral part of daily communication and information exchange tools. Facebook has remained one of the most widely utilized and has 900 million active users globally. In Malaysia, the platform has act as a tool for connection, news, and financial activities. The user base keeps expanding, and individuals increasingly share their personal information online; this will cause increasing cybersecurity threats.

Cybercriminals have exploited the inherent trust frequently, and social media environments are too open to launch malicious activities such as phishing, scams, and social attacks. These threats not just affect user privacy but also can lead to financial and identity losses. Phishing is one of the methods that send fraudulent messages that appear legitimate to harvest sensitive user details. These attacks will often act as fake login pages, friend requests and messages that claim any rewards offering or account verification. When users provide their credentials or finacial details, attackers can use them for any illegal activities.

Research has suggested that social media phishing is more effective due to targeting human psychology. Attackers can use psychological triggers such as fear, curiosity, and urgency to attack. Attackers have been known to exploit high-stress global events, such as sending virus-themed scam messages to make people fear the COVID-19 pandemic. These kinds of tactics will also apply impersonation by pretending victim's family or friend to bypass victim's natural skepticism. By understanding human factors, these socio-cognitive elements will make university students vulnerable. Therefore, effective preventative strategies are critical.

1.2 Problem Statement

Facebook has become one of the main communication tools in the daily life of Generation Z, especially for university students. This demographic will be well-

versed in digital safety given their high frequency of internet usage. The growth of Facebook usage has increase cause growth of phishing, scams and other forms of social engineering cyberthreats across different demographic groups. Research will indicate that phishers increasingly rely on social engineering tactics that target human elements of security.

A significant gap between implementation of preventive measures and their effectiveness among students has existed. Although that government has approached awareness seminars and warning messages on social media platforms, it is still vital to act against phishing and social engineering attacks. Students at university are more open to connect to strangers and less awareness of sophisticated phishing deception. This will cause them to be more likely to be victims of online fraud, even with general preventative measures in place.

This research is to aim to understand the awareness among Universiti Utara Malaysia (UUM) students based on phishing, scam and social engineering. This project will also try to gain understanding of why and how phishing attacks. This project will also attempt to gain understandings of why and how phishing attacks will let students trust strangers and test effectiveness of claimed preventive measures students purportedly untake. By using Protection Motivation Theory (PMT), this study will determine if students' is perceived severity and vulnerability truly translate into effective protective behavior

1.3 Research Questions

1.3.1 What is the level of awareness among UUM students regarding phishing, scams, and social engineering threats on Facebook?

1.3.2 What actions do students of UUM take when they come across phishing, scams, or suspicious messages on Facebook?

1.3.3 What are the most frequent phishing protective measures employed by UUM students when using Facebook?

1.3.4 What Factors will influence UUM students' ability to recognize and respond to phishing, scams, or suspicious messages on Facebook?

1.3.5 What are the ways and types of recommendations that can help Facebook users with awareness and online safety protection enhancement?

1.4 Research Objectives

1.4.1 To study major socio-cognitive factors that enable the occurence of phishing, scam and social engineering attacks against UUM students on Facebook.

1.4.2 To study the reaction followed by UUM students when they have experience with phishing attempts, scams or suspicious messages on Facebook.

1.4.3 To study degree of apprehension among UUM students to potential risk posed by phishing and social engineering at Facebook

1.4.4 To study effectiveness of protective strategies used by UUM students to protective themselves against phishing and scam

1.4.5 To test effect of social engineering strategies on Facebook behavior and security among student of UUM

1.5 Significance of the Study

This research is to focus on understanding how UUM students react and identify phishing threats. The study will focus on identifying key factors that will affect student's awareness and how they react and find out phishing attempts that are related to financial matters. Students can become more aware by identifying these scam patterns when they face online. This study will highlight the importance of promoting safer online practices. This study will also encourage healthy digital behavior, and results will also guide social media platforms to improve their cybersecurity levels for society.

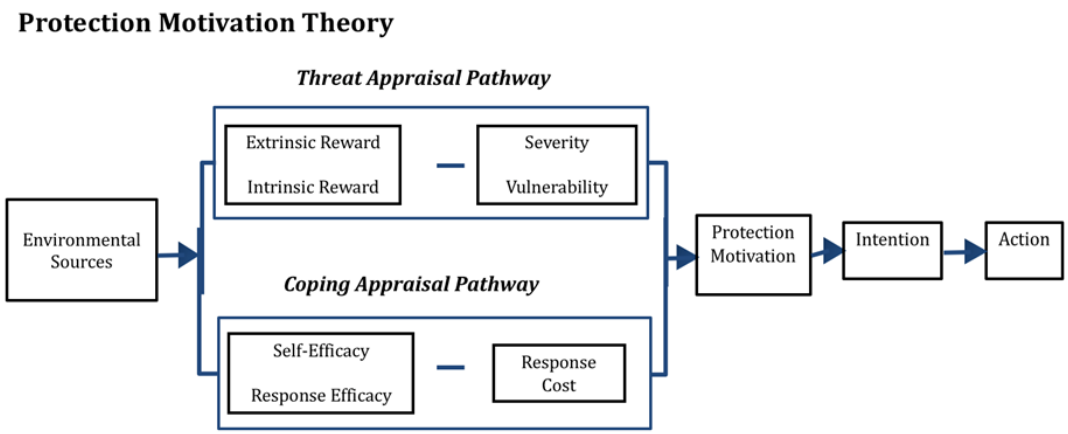

Figure 1: Protection Motivation Theory

1.6 Scope, Limitations & Delimitations of the Study

1.6.1 Scope

This research focuses on awareness of phishing and user behavior among UUM Facebook student users. This study will focus on undergraduate students aged between 18 and 25 years old. This research will utilize questionnaires to distributed to UUM

students for data collection regarding demographics, phishing experiences and awareness levels. The expected sample size is 50 to 100 students, and the scope is strictly confined to behavioral and awareness-related aspects and does not involve experimental interventions.

1.6.2 Limitations

This study has faced several limitations based on the sample population. Firstly, the total number of respondents was 55, which means that the lower end of targeted sample size. Secondly, gender imbalance in data collected which mean that male respondents have 87.3% (48 respondent) and female respondents only have 12.7% (7 respondents). This imbalance has implied that finding has more to male student. Additionally, the study relies on self-reported data, where students may claim to verify links, and 65.5% admitted they don't know how to what steps to do if account is hacked

1.6.3 Delimitations

This study focuses on Facebook only instead of other popular social media platforms. Furthermore, specific focus is on financial phishing and social engineering attacks, it doesnt cover any other cyber threats that don't involve financial and credential theft. The study will also be delimited to the Protection Motivated Theory (PMT) framework to analyze user behavior which means focusing on threat and coping appraisal rather than technical software solutions for phishing prevention.

1.7 Operational Definition of Terms

Phishing: Criminal tracking that will cause victims to expose their personal, financial, and credential data. Usually, links that will connect to legitimate websites. In this study, this refers to methods that are used on Facebook such as fake website pages and friend requests.

Social Engineering: This is the technique where cybercriminals use social media to launch malicious activities. It targets humans with psychological manipulation rather than trying to break software protections.

Protection Motivation Theory (PMT): Theory that identifying individual's motivation to protect themselves from perceived threat.

Threat Appraisal: Component of PMT that will assesses an individual's perception of threat specifically Threat Vulnerability

Coping Appraisal: Component of PMT that assesses individual's ability to deal with threats. This includes belief that protective measures will be effective and confidence in one's ability to execute protective measures.

CHAPTER TWO
LITERATURE REVIEW

2.1 Review of Past Works

2.1.1 Synopsis of Cybersecurity Risks and Social Media

Social media platforms have developed from basic communication tools to essential elements of everyday life in today's digital environment especially for younger people. Facebook is the most popular social media site in the world with billions of active members. With over 23.1 million active users or a sizable share of the population, figures show a strong penetration rate in Malaysia. However, its broad use has made it easier for hackers to thrive. The dangers connected with cybersecurity attacks have increased in direct proportion to the amount of personal, financial and behavioral data that individuals provide online.

According to research by Alkhalil et al. (2021), despite advancements in technology, people are still the weakest link in cybersecurity. Cybercriminals are now more interested in using social engineering to take advantage of human psychology than technical flaws. Phishing is one of the most common and harmful types of cyberattacks among these risks. Because it takes advantage of the openness and trust that are inherent in social networking contexts, phishing on social media is especially successful. Social media phishing in contrast to conventional email phishing could employ peer connections to make fraudulent information seem more authentic.

2.1.2 Facebook Phishing and Social Engineering Strategies

Facebook phishing attempts have evolved from straightforward "Nigerian Prince" frauds to intricate social engineering plans. Social engineering is the practice of coercing people into disclosing private information or doing activities that endanger security.

- Perceived Safety and Social Trust Abuse

The manipulation of social trust is one of the main factors that makes phishing on Facebook successful. Recent research indicates that attackers use social interaction mechanisms like "friend requests," "likes," and "shares" which are intended to promote connection as weapons. In their social circles, users often act under a "perceived safety" bias believing that messages from friends are always secure.

In order to distribute malicious links or requests to the victim's friends list, attackers often breach authentic accounts. Compared to anonymous spam, the message's click-through rate is greater since the receiver is less inclined to closely examine the link because it comes from a known contact. For example, situations in which a hacked account sends a message such as "Is this you in this video?" take advantage of the victim's trust and interest, circumventing their danger assessment systems. This phenomena shows how social trust is a double-edged blade that both lowers barriers against social engineering and facilitates bonding.

- Complexity of Attacks and AI-Powered Deception

  Another important issue in contemporary writing is the intricacy of assaults. Artificial Intelligence (AI) and other cutting-edge technologies are increasingly being used by attackers to create very convincing phishing lures. Scammers may create grammatically accurate and contextually appropriate messages using generative AI that replicate the tone of genuine alerts or friends' writing styles making it challenging for consumers to discern between authentic and fake data.

  Scammers often take use of platform-specific characteristics to increase credibility. This involves creating phony profiles that mimic actual organizations or common friends, as well as manipulating the notification system for example sending fictitious "account suspension" notifications that show up in the user's notification bar). According to Kaspersky (2024), these "realistic" assaults seriously damage a user's sense of self-efficacy since even

watchful users could find it difficult to spot minute differences in URLs or profile information.

2.1.3 Students' Behavior and Psychological Vulnerabilities

University students who are sometimes called "digital natives," are ironically easy targets for these assaults. Despite having a high level of technological proficiency, fraudsters often take advantage of urgency, curiosity and academic pressure to influence their online activity. Even though they are quite tech-savvy, fraudsters often take advantage of their online activity by using things like urgency, curiosity and academic pressure.

- Manipulation of Emotions and Urgency
  Scammers frequently make people feel like they need to act quickly to stop them from thinking critically according to research. Messages that say an account will be canceled or that a scholarship opportunity is ending make people feel something right away. The cognitive load lowers pupils' ability to halt and double-check material when they are under pressure to meet deadlines and deal with academic stress. According to UUM students' qualitative comments, "rushing" and "panic" are the main causes of falling for scams since emotional pressure takes precedence over prior awareness of cybersecurity threats.
- Greed and Financial Fraud
  Scammers still use money as a powerful incentive to target youngsters. "Reward scams," which offer free prizes, scholarships, or easy money make use of the financial difficulties that college students sometimes encounter. According to research, people have a tendency to underestimate the dangers associated with a prospective financial windfall which a habit known as the "optimism bias." This is supported by data from UUM students who report seeing notifications touting incentives or account verifications at high rates.

2.1.4 Effects of Attacks That Are Successful

The negative effects of falling for phishing go beyond short-term financial loss. Several levels of effects are identified by literature:

- **Financial and Data Loss:**
  Unauthorized transactions and identity theft are the most obvious consequences.

- **Academic and Psychological Impact:**
  Following a scam, students may endure extreme stress, anxiety, and distraction from their academic responsibilities. The "response cost" of the time and effort required to recover compromised accounts or dispute financial charges hinders their study.

- **Erosion of Self-Efficacy:**
  Students who are repeatedly exposed to successful scams may develop "learned helplessness," a belief that self-defense is pointless because the assaults are too complex to resist.

2.1.5 Deficits in Current Literature

This study attempts to fill in a number of gaps in the vast body of research on phishing:

- **Context Specificity**
  The majority of research focuses on business settings or regular internet users. Research on Malaysian university students (UUM) and their distinct socio-academic setting is rare.

- **Application of PMT to Social Engineering**
  Although Protection Motivation Theory (PMT) has been used to cybersecurity, there aren't many research that specifically connect PMT to Facebook's "friend impersonation" and "social trust exploitation" vectors.

- **Current Vectors of Attack**
  AI-generated frauds are one example of a rapidly evolving approach that calls for updated study to understand how classic theoretical models survive these new challenges.

2.2 Theoretical & Conceptual Background

2.2.1 Theoretical Background

Protection Motivation Theory (PMT) .This research uses the Protection Motivation Theory (PMT) to comprehend the elements impacting UUM students' cybersecurity knowledge and behavior. PMT was first developed by Rogers in 1975 to explain how individuals respond to fear appeals in health communication. Since then, it has been successfully applied to the area of information systems security. It offers a thorough framework for elucidating the reasons people choose to participate (or refrain from engaging) in protective actions when confronted with a danger.

According to PMT, the desire to defend oneself which known as Security Motivation is the outcome of two cognitive mediating processes which are Threat Appraisal and Coping Appraisal.

- **Threat Assessment (The "Why")**
  Threat appraisal is the process by which a person evaluates the possible risk. "Is this a problem for me?" is addressed by it. There are two important sub-constructs in this process:
  1. **Perceived Severity**
     This is a person's assessment of the seriousness of the repercussions should the danger materialize. In the context of this research, it refers to the degree to which a student fears losing their Facebook account, money, or data privacy. A student's perception of identity theft's seriousness is low if they consider it to be a mere annoyance.
  2. **Perceived Vulnerability**

The term "perceived vulnerability" describes how likely it is that a person will really encounter the hazard. Students may not take action if they think "it won't happen to me" (low vulnerability), even if they are aware that phishing is serious (high severity). When a student admits they may be a target, they have a high perceived vulnerability.

- **Evaluation of Coping (The "How")**

The process of coping appraisal entails the person evaluating their capacity to prevent the danger. It provides a response to the query, "Can I do something about it?" Three essential sub-constructs make up this process:

1. **Response Efficacy**
   This is the assumption that the advised protective activity will truly be helpful in lessening the danger. For example, does the student feel that setting Two-Factor Authentication (2FA) would actually stop a hacker?
2. **Self-Efficacy**
   This is the individual's confidence in their own capacity to perform the prescribed conduct. Although a student feels they lack the technical know-how to put up 2FA, they may think it works (high response efficacy) (low self-efficacy).
3. **Response Cost**
   The perceived costs of the protective activity, such as time, effort, money, or annoyance are referred to as response costs. The person is less likely to engage in the action if the "cost" such as the trouble of checking each link exceeds the apparent gain.

2.2.2 Conceptual Framework

This research builds a conceptual framework based on PMT principles to illustrate the connection between these elements and UUM students' cybersecurity actions.
The framework proposes a linear but intricate connection, as shown in Figure 1 (see the picture in the appendix):

1. Input : The elements of the Threat Appraisal (Severity, Vulnerability) and Coping Appraisal (Response Efficacy, Self-Efficacy, Response Cost) are the independent variables.

2. Process: These factors affect Security Motivation, the mediating variable. This is the student's willingness or internal will to behave.

3. Output: Security Behavior is the last dependent variable. This reflects what students really do such spotting phishing attempts, rejecting dubious friend requests, updating passwords and reporting fraud.

According to the concept, Security Motivation is impacted by both Threat Appraisal and Coping Appraisal. In particular:

1. The perceived requirement for action is increased by both high threat severity and high threat vulnerability.

2. The perceived capacity to act is increased when response efficacy and self-efficacy are both high.

3. The incentive to act is adversely affected by high response costs, which serve as a barrier.

By using this theoretical framework, the study may go beyond only asking "Do students know about phishing?" to comprehend "What motivates students to protect themselves?" It assists in determining if the lack of protection results from a lack of confidence or abilities (poor coping assessment) or a lack of anxiety (low threat appraisal).

2.2.3 Hypotheses

The following theories are suggested to be investigated in this study based on the conceptual framework and the analysis of earlier research:

1. H1: A positive correlation exists between the Perceived Severity of phishing attacks and the self-protective conduct of pupils. Students will behave more defensively if they are more afraid of the repercussions.

2. H2: There is a link between feeling vulnerable to scammers and taking steps to protect yourself. Pupils are more willing to take safeguards if they feel personally at danger.

3. H3:Self-Efficacy and self-protective conduct are positively correlated.Students are more likely to adopt safe practices if they have faith in their capacity to spot scams.

4. H4: There is a favorable correlation between Response Efficacy and self-protective behavior. Students are more likely to employ security measures if they have faith in their effectiveness.

5. H5: Response Cost has a detrimental effect on self-protective behavior. Students who see security measures as excessively time-consuming or challenging are less inclined to implement them.

A better comprehension of the psychological factors influencing Facebook's cybersecurity policies is made possible by this framework and the related hypotheses which provide a solid foundation for evaluating the quantitative data gathered from UUM students.

CHAPTER THREE
METHODOLOGY

3.1 Research Design

This research is given a quantitative method of research with the use of the cross-sectional survey. The quantitative approach was chosen due to the possibility of measuring the level of student awareness numerically and conducting statistical tests to prove the relationships between variables based on the Protection Motivation Theory (PMT). A quantitative design would be more appropriate compared to qualitative methods used in investigating reasons and motivations behind certain phenomena since it would allow concluding larger patterns and generalizing the results on the whole population of Universiti Utara Malaysia (UUM) students.

The design of the research is cross-sectional which implies that the data are gathered on the target population at a particular time and not over a span of time. Such a method is quite beneficial towards this research because of time limitations and necessity to represent a snapshot of existing cybersecurity situation among students. It enables one to easily determine the current perception of threats like phishing, scams, and social engineering on Facebook by students.

The underpinning of the framework of this research lies in the Protection Motivation Theory (PMT) which states that the intention to adopt a protective behavior by an individual is acquired through two primary cognitive processes; Threat Appraisal and Coping Appraisal.

- Threat Appraisal: This element of the design assesses how much the students perceive the threat (i.e. financial loss, identity theft) to be serious, and their perceived susceptibility (vulnerability to becoming a victim).

- Coping Appraisal: This construct measures how the students believe that the given response will work (Response Efficacy) and how they are confident that they will be able to perform the given response (Self-Efficacy).

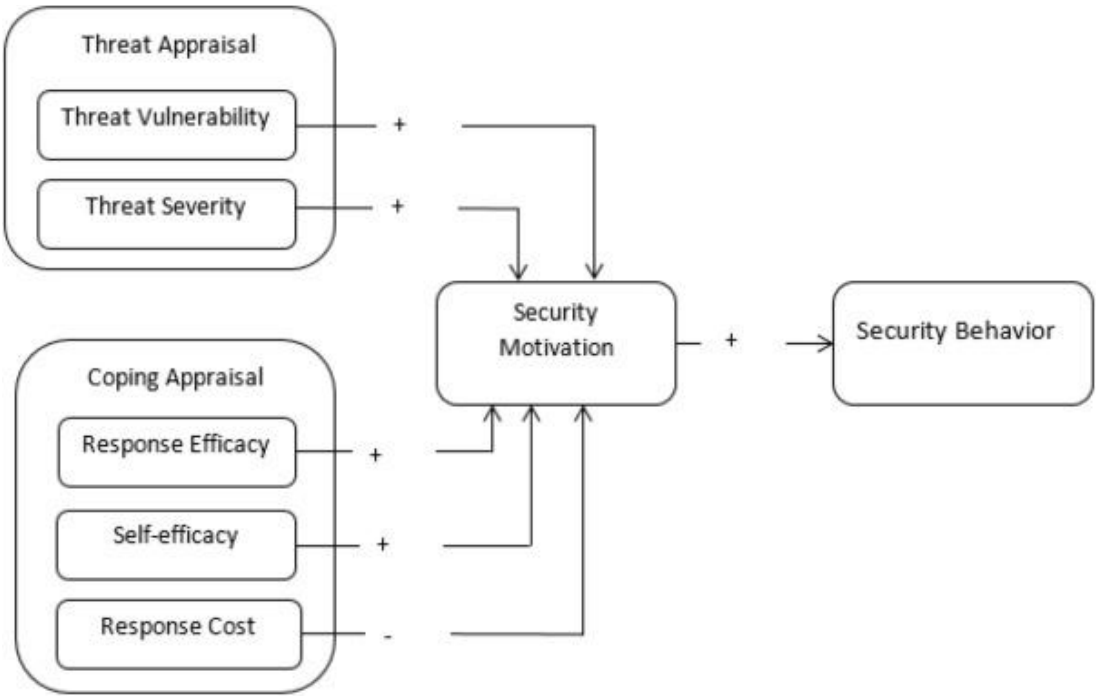

Figure 2 2: Protection Motivation Theory Framework

Incorporating the theoretical constructs into a correlational research design the study will test special assumptions whether the elevation of perceived threat and efficacy results in a greater adoption of security measures like the use of two-factor authentication or checking suspicious links.

3.2 Instrument Development

The major research tool of this study is a questionnaire. The creation of this tool was a three stage process: generation of items, content validation and refinement.

3.2.1 Item Generation and Operationalization

Questionnaire items were created by modifying those who came out of earlier validated scales of other cybersecurity studies and customise them to the area of Facebook use among university students. A total of eight sections is assigned to the instrument to cover the whole data:

- **Demographics (Section 1 and 2):** The sections will filter the respondents into being up-to-date students of UUM with active Facebook accounts. The demographic trends are analyzed by gathering the data based on age, gender, and frequency of use.
- **Threat Appraisal Constructs (Section 3):** The items in this section assess Abuse of Social Trust and the Perceived Severity. As an example, items determine whether students trust links of their friends blindly or they fear to lose money.
- **Coping Appraisal Constructs (Section 3):** Items here assess Self-Efficacy (i.e., confidence in ability to notice fake login pages) and Response Efficacy (i.e., belief in the effectiveness of reporting scams to the authorities).
- **Protective Behavior (Section 4 & 7):** This section operationalizes the dependent variable in which respondents will be asked about particular behaviors such as changing passwords and using two-factor authentication.
- **Experience and Qualitative Assessment (Sections 5, 6 and 8):** These parts get the data about particular scam types experienced (e.g., fake rewards) as well as open-ended questions to explore it further.

The core PMT constructs were selected using a 7-point Likert scale (1=Strongly Disagree, 7=Strongly Agree). Included in this scale are more precise perspectives than that of a normal 5-point scale, and a student attitude can slightly tilt with this scale to either agree or disagree with it.

3.2.2 Content Validity Index (CVI)

To determine that the instrument measures what it was designed to measure, the content validity analysis was done. The draft questionnaire was checked by a panel of three experts Muhammad Farhan bin Ismail, Lee Zhi Xuen and Nur Aisyah binti Rahman.

The professionals evaluated the items on a relevance scale of 4 (1 = Not Relevant, 4 = Highly Relevant). According to these ratings, Content Validity Index (CVI) was determined:

- **Item-CVI (I-CVI):** This was obtained as the number of experts who rated 3 or 4 as a proportion of the total number of experts. In the case of the section (Phishing & Social Engineering Factors), all the experts rated it 4, which led to I-CVI of 1.00.

- **Scale-CVI (S-CVI):** The mean of the I-CVI scores of the whole tool was calculated. The validation process validated the fact that the items on the factor of Protective Behaviour and Perceived Severity were very inherent to the study objectives. As such, any proposed items were to be held to the end distribution.

3.3 Data Collection Procedure

The process of data collection was implemented in a systematic manner to have a sample that would represent the student population of the UUM and at the same time, follow the ethical research practices.

3.3.1 Pre-Data Collection

The process of data collection was implemented in a systematic manner to have a sample that would represent the student population of the UUM and at the same time, follow the ethical research practices.

3.3.2 Sampling and Distribution

Sampling and Distribution Convenience sampling method was used in the study. Although this is none-probability type of sampling method, it was considered to be appropriate given the limitation of the academic semester and the necessary targeting of a certain and accessible population.

- **Target Population:** The targeted respondents included the undergraduate students of Universiti Utara Malaysia who actively use Facebook.

- **Dissemination:** The survey link was shared on high traffic online platforms visited by the students. This consisted of WhatsApp based course discussion groups, residential hall (Inasis) based chat groups and social media based UUM student community pages.

- **Duration:** Data collection was kept open to ensure that response is adequate, as it took a duration of two weeks.

3.3.3 Ehical Considerations

There were no ethical issues that were not observed during the process. At the start of the Google Form, a compulsory consent section was included. This part helped the participants of the study to understand the academic intent of the study and ensured the anonymity of their information. It made it clear that there would be no publicly disclosed personally identifiable information (PII): e.g., names or matric numbers. Moreover, procedural precautions were taken in the form of screening questions, which allowed the removal of non-students or non-active Facebook users of the dataset to complete the validity of the final sample.

3.4 Data Analysis Procedure

When the data collection part was over, the data was pulled out of Google Forms and then imported to Microsoft Excel to be processed. Analysis was done in a three-step numerical workflow: Data Cleaning, Descriptive Analysis, and Hypothesis Testing.

3.4.1 Data Cleaning

The raw data were initially checked in errors. This included deleting half-responses, as well as ensuring that all the respondents qualified (UUM student status). Validity of any responses that depicted straight-lining patterns (e.g. the response of choosing 3 to all questions) was checked.

3.4.2 Descriptive Statistics

Descriptive analysis was conducted to obtain the results of the demographic profile of the respondents. Categorical variables were frequency distributed and percentaged including age, gender and frequency frequency of Facebook use. This measure gave a clear understanding of the sample features as this ensured that the respondents were mostly young adults between the ages of 18-23 years and use Facebook regularly. Also, the PMT constructs were analyzed with central tendency/mean and dispersion/standard deviation that has been used to establish the average degree of awareness and the self-efficacy levels of the students.

3.4.3 Testing Research Hypotheses

Inferential statistical analysis was done to answer the research objectives, as well as to test the proposed hypotheses (H1 through H5).

- **Pearson Correlation Coefficient:** This test was to test the strength as well as direction of the relationship between the independent variables (Perceived Severity, Vulnerability, Self-Efficacy, Response Efficacy) and the dependent variable (Protective Behavior). As an illustration, to test H3 (Increased self-efficacy should be reflected in terms of self-protective behaviour), Self-Efficacy score and the Protective Behavior score were correlated.

- **Thematic Analysis:** Thematic Analysis was applied to the qualitative questions in Section 8, which were open-ended. Feedback by students concerning reasons behind their fall into scams were labeled into themes like Emotional Urgency, Trust in Friends and the lack of Technical Knowledge.

CHAPTER FOUR
ANALYSIS AND FINDINGS

4.1 Descriptive Statistics of Respondents

This section provides the description of the demographic profile of the respondents. Originally 55 responses were obtained. Following the data cleaning activity, which was described in Chapter 3 (Section 3.4.1), 4 respondents were eliminated because they were not able to conform to the screening criteria (a student of UUM who answers No to the question). This gave a final sample size of 51.

Demographic Variable	Category	Frequency (n)	Percentage (%)
Student Status	UUM Student	51	100.00%
Facebook Account	Active User	51	100.00%
Gender	Male	45	88.20%
	Female	6	11.80%
Age Group	18 – 20 years	38	74.50%
	21 – 23 years	13	25.50%
	24 years and above	0	0.00%
Facebook Usage	Daily	45	88.20%
	Several times a week	4	7.80%
	Rarely	2	3.90%

Table 1: Demographic Profile of Respondents (N=51)

4.1.1 Gender and Age Distribution Gender:

There is a substantial gap in gender distribution, whereby, the student population was composed mainly of male students (88.2) 45, and female students (11.8) 6. On the age, most of the respondents are in the age bracket of 18 to 20 years (74.5), then the 21 to 23 years (25.5). The final valid sample was only made up of no respondents that had

an age of 24 years or more. This shows that the study mainly represents the opinion of the younger undergraduate students.

4.1.2 Facebook Usage Habits

The screening tool allowed to make sure 100 percent of the respondents have an active Facebook account. The information on the number of usages shows that UUM students are very active on the platform. Preponderant 88.2 (n=45) used facebook on a daily basis, with 7.8% of respondents using frequently between several times weekly. Only a minor group (3.9%) stated that they used the platform Rarely. Such a high rate of usage proves the topicality of the study since daily users are more likely to be exposed to the risks of phishing and social-engagement.

4.2 Data Pre-Processing

To guarantee the integrity and quality of the dataset, data pre-processing was carried out before to the primary statistical analysis. After being exported from Google Forms in CSV format, the raw data was entered into the analysis program. The actions listed below were taken:

i. **Data Cleaning:** Incomplete answers, duplicate entries, and straight-lining behavior in which respondents choose the same response for every question were examined in the dataset. Since every question had to be answered according to the survey conditions, no significant missing data were discovered.
ii. **Screening:** The questions about being a UUM student and having an active Facebook account were examined. Since all 55 replies satisfied the requirements of being active Facebook users within the academic setting, they were all considered legitimate for study.
iii. **Variable Coding:** To make statistical calculations easier, the Likert scale items, which range from 1 = Strongly Disagree to 7 = Strongly Agree were numerically coded.

iv. **Calculate Variables:** By averaging the items associated with each of the five primary constructs Perceived Severity, Perceived Vulnerability, Self-Efficacy, Response Efficacy, and Protective Behavior composite scores were computed.

Following these steps, N = 55 was the final sample size, which was used to all further studies.

4.3 Main Data Analysis

This section presents the analysis of the core research variables: Phishing Factors such as Awareness and Threat Perception and Protective Behaviour. A descriptive analysis was conducted based on the distribution of mean scores and response frequencies.

A distinct pattern was observed across the dataset where a significant majority of respondents selected "1 – Strongly Disagree" for most questions. It is crucial to interpret this correctly, as the meaning changes depending on the phrasing of the question:

Positive Outcome: When respondents "Strongly Disagreed" with risky behaviors such as "I trust links from friends", it indicates that they have a high level of security awareness.

Negative Outcome: When respondents "Strongly Disagreed" with safety habits such as "I frequently change my password", it indicates poor security practices.

4.3.1 Phishing & Social Engineering Awareness

The first objective was to determine the factors influencing awareness, specifically regarding social trust.

Trust in Unverified Links: The data indicates that UUM students have a high level of skepticism. When asked if they "usually trust links sent by Facebook friends without verifying them," 69.1% of respondents selected "Strongly Disagree." This

shows that students generally understand the concept of social engineering and do not blindly trust content based solely on friendship.

Perception of Scam: Interestingly, 67.3% of students "Strongly Disagreed" with the statement that said phishing messages are unrealistic and easy to identify. This means that students recognize that modern scams are sophisticated and realistic, making them difficult to detect.

4.3.2 Analysis of Threat Appraisal

The data indicates that UUM students possess a high level of threat assessment ability. They are acutely aware of the risks with Facebook scams.

Perceived Severity: When presented with the statement "Falling victim to a financial scam on Facebook would not cause me significant stress," a dominant 69.1% of respondents Strongly Disagreed. This confirms that students believe the consequences of scams as severe and highly stressful.

Perceived Vulnerability: Similarly, regarding vulnerability, 67.3% Strongly Disagreed with the statement "I believe that I am not personally vulnerable to identity theft". This demonstrates that students do not have an optimism bias, they realistically acknowledge that they are potential targets.

4.3.3 Coping Appraisal

Despite their high awareness of the risks, students lack confidence in their Self-Efficacy.

Low Self-Efficacy: A significant gap was found in the students' technical confidence. 67.3% Strongly Disagreed that they find it easy to distinguish between real and fake notifications. Furthermore, 65.5% of students admitted that they do not know what specific steps to take if their account is hacked. This indicates that while students are scared of scams, they have only limited specific skills to identify and prevent it.

Mixed Response Efficacy: Students believe that personal verification is effective (65.5% disagreed that verifying URLs is ineffective), which means they know that checking link is works. However, they lack confidence in the platform's tools; 65.5% Strongly Disagreed that reporting suspicious accounts to Facebook actually helps reduce scams.

Analysis: This creates a worrying gap: students know they are targets, but do not believe they have the skills to identify the attacks.

4.3.4 Protective Behaviour

The analysis of protective behaviors reveals a mix of passive and active defense strategies.

Passive vs. Active Security: There is a strong disconnect between intent and action. While 67.3% claimed that they will verify the links before clicking, a contrasting 70.9% "Strongly Disagreed" that they frequently change their Facebook passwords.

Passive response: The data show that students tend to respond passively rather than proactively prevent incidents. They are likely to block a sender after an incident occurs, but they rarely perform routine maintenance like changing their Facebook password.

CHAPTER FIVE
DISCUSSION AND CONCLUSION

5.1 Discussion of the Findings

This study aimed to identify the factors of phishing and social engineering attacks among UUM students on Facebook. Based on the Protection Motivation Theory (PMT) framework outlined in the proposal, the findings confirm three critical factors:

5.1.1 Factor 1: Abuse of Social Trust

Finding:

Quantitative Data: The survey showed that 69.1% of students claimed they do not trust unverified links from friends.

Qualitative Data: However, in the open-ended responses, students admitted that "Trust in friends" was one of the main reasons they fall for scams, often lowering their guard when a message appears to come from a known contact.

Conclusion: While students theoretically know they shouldn't trust links, in actual attacks, the social pressure and trust factor will override this awareness.

5.1.2 Factor 2: Attack Complexity vs. Low Self-Efficacy

Findings: A majority of students (67.3%) admitted they cannot easily distinguish between real and fake Facebook notifications, and 67.3% do not feel confident of recognizing fake login pages.

Discussion: This confirms that the high complexity of modern social engineering attacks has outpaced the technical capabilities of students. They know the attacks exist, but they lack the technical ability to identify them.

5.1.3 Factor 3: High Response Cost

Findings: The data supports this. While blocking a sender is common (90.9%), proactive measures that require effort, specifically "Changing Facebook Password", are rarely used, 70.9% of students said that they are not doing it frequently.

Discussion: This indicates that "Response Cost" is a major factor preventing students from developing good safety habits, they prefer convenient and passive measures over proactive but not time-consuming ones.

5.2 Contributions of the Study

This research contributes to the existing body of knowledge in two specific ways:

Validating Protection Motivation Theory (PMT) for UUM Students: The study confirms that the Protection Motivation Theory is applicable to UUM students. The study specifically highlights that Self-Efficacy is the weakest link in the student's defense mechanism, rather than a lack of general self-awareness.

Identifying the Urgency Variable: Beyond the proposed factors, this study found that Urgency or Panic, such as fear of account suspension is a critical psychological factor that scammers use to bypass a student's logical reasoning.

5.3 Limitations and Directions/Suggestions for Future Research

Limitations:

Sample Demographics: The study sample was heavily male dominated (87.3%). This may limit the generalizability of the findings regarding "Risk Perception," as male and female users often exhibit different online risk behaviors.

Response Bias: A "straight lining" pattern was observed in the data, suggesting some respondents may have rushed through the survey, consistent with the “rush” factor identified in the qualitative feedback.

Suggestions for Future Research:

Focus on Balanced Gender: Future studies should aim for a more balanced gender ratio to understand whether female students at UUM exhibit different factors.

Experimental Methodology: Future researchers should not simply ask students if they can identify scams, but should conduct "simulated phishing activities" to observe how many students would click on fake links in real-time to determine the real factor why students fall for scams.

5.4 Conclusion

This research successfully identified the main factors of phishing and social engineering attacks among UUM students on Facebook. The study concludes that the main factors are not a lack of awareness, but rather Abuse of Social Trust, High Attack Complexity causing low self-confidence, and the Psychological Manipulation of urgency.

Data shows that while UUM students are aware of their vulnerability, they lack the ability to identify sophisticated cyberattacks. Therefore, the solution is not just raising awareness but also in providing skills training. Specifically, this involves teaching students how to verify a link and how to identify fake login pages to enhance their self-protection and reduce the risk of falling into scams.

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APPENDICES

Appendix A
CVI Form